Ⓑ On a scale of 1–10, how would you rate your mastery of this section in light of your responses on the checklist? How can you improve this?

7.5 Systems of Measurement

Learning Objectives

By the end of this section, you will be able to:

- Make unit conversions in the U.S. system
- Use mixed units of measurement in the U.S. system
- Make unit conversions in the metric system
- Use mixed units of measurement in the metric system
- Convert between the U.S. and the metric systems of measurement
- Convert between Fahrenheit and Celsius temperatures

☑ **BE PREPARED** 7.13 Before you get started, take this readiness quiz.

Multiply: $4.29(1000)$.
If you missed this problem, review Example 5.18.

☑ **BE PREPARED** 7.14 Simplify: $\frac{30}{54}$.
If you missed this problem, review Example 4.20.

☑ **BE PREPARED** 7.15 Multiply: $\frac{7}{15} \cdot \frac{25}{28}$.
If you missed this problem, review Example 4.27.

In this section we will see how to convert among different types of units, such as feet to miles or kilograms to pounds. The basic idea in all of the unit conversions will be to use a form of 1, the multiplicative identity, to change the units but not the value of a quantity.

Make Unit Conversions in the U.S. System

There are two systems of measurement commonly used around the world. Most countries use the metric system. The United States uses a different system of measurement, usually called the U.S. system. We will look at the U.S. system first.

The U.S. system of measurement uses units of inch, foot, yard, and mile to measure length and pound and ton to measure weight. For capacity, the units used are cup, pint, quart and gallons. Both the U.S. system and the metric system measure time in seconds, minutes, or hours.

The equivalencies among the basic units of the U.S. system of measurement are listed in Table 7.2. The table also shows, in parentheses, the common abbreviations for each measurement.

U.S. System Units	
Length	**Volume**
1 foot (ft) = 12 inches (in) 1 yard (yd) = 3 feet (ft) 1 mile (mi) = 5280 feet (ft)	3 teaspoons (t) = 1 tablespoon (T) 16 Tablespoons (T) = 1 cup (C) 1 cup (C) = 8 fluid ounces (fl oz) 1 pint (pt) = 2 cups (C) 1 quart (qt) = 2 pints (pt) 1 gallon (gal) = 4 quarts (qt)

Table 7.2

U.S. System Units	
Weight	**Time**
1 pound (lb) = 16 ounces (oz) 1 ton = 2000 pounds (lb)	1 minute (min) = 60 seconds (s) 1 hour (h) = 60 minutes (min) 1 day = 24 hours (h) 1 week (wk) = 7 days 1 year (yr) = 365 days

Table 7.2

In many real-life applications, we need to convert between units of measurement. We will use the identity property of multiplication to do these conversions. We'll restate the Identity Property of Multiplication here for easy reference.

$$\text{For any real number } a, \qquad a \cdot 1 = a \qquad 1 \cdot a = a$$

To use the identity property of multiplication, we write 1 in a form that will help us convert the units. For example, suppose we want to convert inches to feet. We know that 1 foot is equal to 12 inches, so we can write 1 as the fraction $\frac{1 \text{ ft}}{12 \text{ in}}$. When we multiply by this fraction, we do not change the value but just change the units.

But $\frac{12 \text{ in}}{1 \text{ ft}}$ also equals 1. How do we decide whether to multiply by $\frac{1 \text{ ft}}{12 \text{ in}}$ or $\frac{12 \text{ in}}{1 \text{ ft}}$? We choose the fraction that will make the units we want to convert *from* divide out. For example, suppose we wanted to convert 60 inches to feet. If we choose the fraction that has inches in the denominator, we can eliminate the inches.

$$60 \cancel{\text{ in}} \cdot \frac{1 \text{ ft}}{12 \cancel{\text{ in}}} = 5 \text{ ft}$$

On the other hand, if we wanted to convert 5 feet to inches, we would choose the fraction that has feet in the denominator.

$$5 \text{ ft} \cdot \frac{12 \text{ in}}{1 \cancel{\text{ ft}}} = 60 \text{ in}$$

We treat the unit words like factors and 'divide out' common units like we do common factors.

HOW TO

Make unit conversions.

Step 1. Multiply the measurement to be converted by 1; write 1 as a fraction relating the units given and the units needed.
Step 2. Multiply.
Step 3. Simplify the fraction, performing the indicated operations and removing the common units.

EXAMPLE 7.44

Mary Anne is 66 inches tall. What is her height in feet?

Solution

Convert 66 inches into feet.

Multiply the measurement to be converted by 1.	$66 \text{ inches} \cdot 1$
Write 1 as a fraction relating the units given and the units needed.	$66 \text{ inches} \cdot \frac{1 \text{ foot}}{12 \text{ inches}}$

Multiply.	$\frac{66 \text{ inches} \cdot 1 \text{ foot}}{12 \text{ inches}}$
Simplify the fraction.	$\frac{66 \cancel{\text{ inches}} \cdot 1 \text{ foot}}{12 \cancel{\text{ inches}}}$
	$\frac{66 \text{ feet}}{12}$
	5.5 feet

Notice that the when we simplified the fraction, we first divided out the inches.

Mary Anne is 5.5 feet tall.

TRY IT 7.87 Lexie is 30 inches tall. Convert her height to feet.

TRY IT 7.88 Rene bought a hose that is 18 yards long. Convert the length to feet.

When we use the Identity Property of Multiplication to convert units, we need to make sure the units we want to change from will divide out. Usually this means we want the conversion fraction to have those units in the denominator.

EXAMPLE 7.45

Ndula, an elephant at the San Diego Safari Park, weighs almost 3.2 tons. Convert her weight to pounds.



Figure 7.5 (credit: Guldo Da Rozze, Flickr)

Solution

We will convert 3.2 tons into pounds, using the equivalencies in Table 7.2. We will use the Identity Property of Multiplication, writing 1 as the fraction $\frac{2000 \text{ pounds}}{1 \text{ ton}}$.

	3.2 tons
Multiply the measurement to be converted by 1.	$3.2 \text{ tons} \cdot 1$
Write 1 as a fraction relating tons and pounds.	$3.2 \text{ tons} \cdot \frac{2000 \text{ lbs}}{1 \text{ ton}}$
Simplify.	$\frac{3.2 \cancel{\text{ tons}} \cdot 2000 \text{ lbs}}{1 \cancel{\text{ ton}}}$
Multiply.	6400 lbs
	Ndula weighs almost 6,400 pounds.

TRY IT 7.89 Arnold's SUV weighs about 4.3 tons. Convert the weight to pounds.

TRY IT 7.90 A cruise ship weighs 51,000 tons. Convert the weight to pounds.

Sometimes to convert from one unit to another, we may need to use several other units in between, so we will need to multiply several fractions.

EXAMPLE 7.46

Juliet is going with her family to their summer home. She will be away for 9 weeks. Convert the time to minutes.

Solution

To convert weeks into minutes, we will convert weeks to days, days to hours, and then hours to minutes. To do this, we will multiply by conversion factors of 1.

	9 weeks
Write 1 as $\frac{7 \text{ days}}{1 \text{ week}}, \frac{24 \text{ hours}}{1 \text{ day}}, \frac{60 \text{ minutes}}{1 \text{ hour}}$.	$\frac{9 \text{ wk}}{1} \cdot \frac{7 \text{ days}}{1 \text{ wk}} \cdot \frac{24 \text{ hr}}{1 \text{ day}} \cdot \frac{60 \text{ min}}{1 \text{ hr}}$
Cancel common units.	$\frac{9 \cancel{\text{wk}}}{1} \cdot \frac{7 \cancel{\text{days}}}{1 \cancel{\text{wk}}} \cdot \frac{24 \cancel{\text{hr}}}{1 \cancel{\text{day}}} \cdot \frac{60 \text{ min}}{1 \cancel{\text{hr}}}$
Multiply.	$\frac{9 \cdot 7 \cdot 24 \cdot 60 \text{ min}}{1 \cdot 1 \cdot 1 \cdot 1} = 90,720 \text{ min}$
	Juliet will be away for 90,720 minutes.

TRY IT 7.91 The distance between Earth and the moon is about 250,000 miles. Convert this length to yards.

TRY IT 7.92 A team of astronauts spends 15 weeks in space. Convert the time to minutes.

EXAMPLE 7.47

How many fluid ounces are in 1 gallon of milk?



Figure 7.6 (credit: www.bluewaikiki.com, Flickr)

Solution

Use conversion factors to get the right units: convert gallons to quarts, quarts to pints, pints to cups, and cups to fluid ounces.

	1 gallon
Multiply the measurement to be converted by 1.	$\frac{1\text{ gal}}{1}\cdot\frac{4\text{ qt}}{1\text{ gal}}\cdot\frac{2\text{ pt}}{1\text{ qt}}\cdot\frac{2\text{ C}}{1\text{ pt}}\cdot\frac{8\text{ fl oz}}{1\text{ C}}$
Simplify.	$\frac{1\text{ gal}}{1}\cdot\frac{4\text{ qt}}{1\text{ gal}}\cdot\frac{2\text{ pt}}{1\text{ qt}}\cdot\frac{2\text{ C}}{1\text{ pt}}\cdot\frac{8\text{ fl oz}}{1\text{ C}}$
Multiply.	$\frac{1\cdot4\cdot2\cdot2\cdot8\text{ fl oz}}{1\cdot1\cdot1\cdot1\cdot1}$
Simplify.	128 fluid ounces
	There are 128 fluid ounces in a gallon.

TRY IT 7.93 How many cups are in 1 gallon?

TRY IT 7.94 How many teaspoons are in 1 cup?

Use Mixed Units of Measurement in the U.S. System

Performing arithmetic operations on measurements with mixed units of measures requires care. Be sure to add or subtract like units.

EXAMPLE 7.48

Charlie bought three steaks for a barbecue. Their weights were 14 ounces, 1 pound 2 ounces, and 1 pound 6 ounces. How many total pounds of steak did he buy?

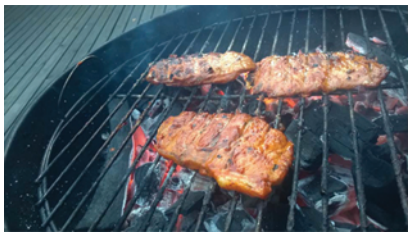


Figure 7.7 (credit: Helen Penjam, Flickr)

Solution

We will add the weights of the steaks to find the total weight of the steaks.

Add the ounces. Then add the pounds.	14 ounces 1 pound 2 ounces + 1 pound 6 ounces 2 pounds 22 ounces
Convert 22 ounces to pounds and ounces.	

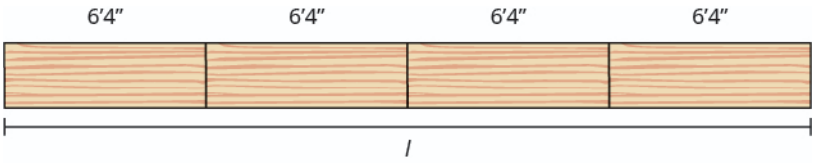
Add the pounds.	2 pounds + 1 pound, 6 ounces 3 pounds, 6 ounces
	Charlie bought 3 pounds 6 ounces of steak.

TRY IT 7.95 Laura gave birth to triplets weighing 3 pounds 12 ounces, 3 pounds 3 ounces, and 2 pounds 9 ounces. What was the total birth weight of the three babies?

TRY IT 7.96 Seymour cut two pieces of crown molding for his family room that were 8 feet 7 inches and 12 feet 11 inches. What was the total length of the molding?

EXAMPLE 7.49

Anthony bought four planks of wood that were each 6 feet 4 inches long. If the four planks are placed end-to-end, what is the total length of the wood?



Solution

We will multiply the length of one plank by 4 to find the total length.

Multiply the inches and then the feet.	$\begin{array}{rr} 6 \text{ feet} & 4 \text{ inches} \\ \times & 4 \\ \hline 24 \text{ feet} & 16 \text{ inches} \end{array}$
Convert 16 inches to feet.	24 feet + 1 foot 4 inches
Add the feet.	25 feet 4 inches
	Anthony bought 25 feet 4 inches of wood.

TRY IT 7.97 Henri wants to triple his spaghetti sauce recipe, which calls for 1 pound 8 ounces of ground turkey. How many pounds of ground turkey will he need?

TRY IT 7.98 Joellen wants to double a solution of 5 gallons 3 quarts. How many gallons of solution will she have in all?

Make Unit Conversions in the Metric System

In the metric system, units are related by powers of 10. The root words of their names reflect this relation. For example, the basic unit for measuring length is a meter. One kilometer is 1000 meters; the prefix *kilo-* means thousand. One centimeter is $\frac{1}{100}$ of a meter, because the prefix *centi-* means one one-hundredth (just like one cent is $\frac{1}{100}$ of one dollar).

The equivalencies of measurements in the metric system are shown in Table 7.3. The common abbreviations for each measurement are given in parentheses.

Metric Measurements		
Length	**Mass**	**Volume/Capacity**
1 kilometer (km) = 1000 m 1 hectometer (hm) = 100 m 1 dekameter (dam) = 10 m 1 meter (m) = 1 m 1 decimeter (dm) = 0.1 m 1 centimeter (cm) = 0.01 m 1 millimeter (mm) = 0.001 m	1 kilogram (kg) = 1000 g 1 hectogram (hg) = 100 g 1 dekagram (dag) = 10 g 1 gram (g) = 1 g 1 decigram (dg) = 0.1 g 1 centigram (cg) = 0.01 g 1 milligram (mg) = 0.001 g	1 kiloliter (kL) = 1000 L 1 hectoliter (hL) = 100 L 1 dekaliter (daL) = 10 L 1 liter (L) = 1 L 1 deciliter (dL) = 0.1 L 1 centiliter (cL) = 0.01 L 1 milliliter (mL) = 0.001 L
1 meter = 100 centimeters 1 meter = 1000 millimeters	1 gram = 100 centigrams 1 gram = 1000 milligrams	1 liter = 100 centiliters 1 liter = 1000 milliliters

Table 7.3

To make conversions in the metric system, we will use the same technique we did in the U.S. system. Using the identity property of multiplication, we will multiply by a conversion factor of one to get to the correct units.

Have you ever run a 5 k or 10 k race? The lengths of those races are measured in kilometers. The metric system is commonly used in the United States when talking about the length of a race.

EXAMPLE 7.50

Nick ran a 10-kilometer race. How many meters did he run?



Figure 7.8 (credit: William Warby, Flickr)

Solution

We will convert kilometers to meters using the Identity Property of Multiplication and the equivalencies in Table 7.3.

	10 kilometers
Multiply the measurement to be converted by 1.	$10\text{ km} \cdot 1$
Write 1 as a fraction relating kilometers and meters.	$10\text{ km} \cdot \frac{1000\text{ m}}{1\text{ km}}$
Simplify.	$\frac{10\cancel{\text{ km}} \cdot 1000\text{ m}}{1\cancel{\text{ km}}}$
Multiply.	10,000 m
	Nick ran 10,000 meters.

TRY IT 7.99 Sandy completed her first 5-km race. How many meters did she run?

TRY IT 7.100 Herman bought a rug 2.5 meters in length. How many centimeters is the length?

EXAMPLE 7.51

Eleanor's newborn baby weighed 3200 grams. How many kilograms did the baby weigh?

Solution

We will convert grams to kilograms.

	3200 grams
Multiply the measurement to be converted by 1.	$3200\ \text{g} \cdot 1$
Write 1 as a fraction relating kilograms and grams.	$3200\ \text{g} \cdot \frac{1\ \text{kg}}{1000\ \text{g}}$
Simplify.	$3200\ \cancel{\text{g}} \cdot \frac{1\ \text{kg}}{1000\ \cancel{\text{g}}}$
Multiply.	$\frac{3200\ \text{kilograms}}{1000}$
Divide.	3.2 kilograms
	The baby weighed 3.2 kilograms.

TRY IT 7.101 Kari's newborn baby weighed 2800 grams. How many kilograms did the baby weigh?

TRY IT 7.102 Anderson received a package that was marked 4500 grams. How many kilograms did this package weigh?

Since the metric system is based on multiples of ten, conversions involve multiplying by multiples of ten. In Decimal Operations, we learned how to simplify these calculations by just moving the decimal.

To multiply by $10, 100,$ or 1000, we move the decimal to the right $1, 2,$ or 3 places, respectively. To multiply by $0.1, 0.01,$ or 0.001 we move the decimal to the left $1, 2,$ or 3 places respectively.

We can apply this pattern when we make measurement conversions in the metric system.

In Example 7.51, we changed 3200 grams to kilograms by multiplying by $\frac{1}{1000}$ $(\text{or } 0.001)$. This is the same as moving the decimal 3 places to the left.

$3200 \cdot \frac{1}{1000}$ $\qquad$ 3200.

3.2 $\qquad$ 3.2

EXAMPLE 7.52

Convert:

(a) 350 liters to kiloliters (b) 4.1 liters to milliliters.

Solution

(a) We will convert liters to kiloliters. In Table 7.3, we see that $1 \text{ kiloliter} = 1000 \text{ liters}$.

	350 L
Multiply by 1, writing 1 as a fraction relating liters to kiloliters.	$350\text{ L} \cdot \frac{1\text{ kL}}{1000\text{ L}}$
Simplify.	$350\cancel{\text{L}} \cdot \frac{1\text{ kL}}{1000\cancel{\text{L}}}$
Move the decimal 3 units to the left.	$350\cancel{\text{L}} \cdot \frac{1\text{ kL}}{1000\cancel{\text{L}}}$
	0.35 kL

ⓑ We will convert liters to milliliters. In Table 7.3, we see that $1 \text{ liter} = 1000 \text{ milliliters}$.

	4.1 L
Multiply by 1, writing 1 as a fraction relating milliliters to liters.	$4.1\text{ L} \cdot \frac{1000\text{ mL}}{1\text{ L}}$
Simplify.	$4.1\cancel{\text{L}} \cdot \frac{1000\text{ mL}}{1\cancel{\text{L}}}$
Move the decimal 3 units to the left.	4.100 mL
	4100 mL

TRY IT 7.103 Convert: ⓐ 7.25 L to kL ⓑ 6.3 L to mL.

TRY IT 7.104 Convert: ⓐ 350 hL to L ⓑ 4.1 L to cL.

Use Mixed Units of Measurement in the Metric System

Performing arithmetic operations on measurements with mixed units of measures in the metric system requires the same care we used in the U.S. system. But it may be easier because of the relation of the units to the powers of 10. We still must make sure to add or subtract like units.

EXAMPLE 7.53

Ryland is 1.6 meters tall. His younger brother is 85 centimeters tall. How much taller is Ryland than his younger brother?

Solution

We will subtract the lengths in meters. Convert 85 centimeters to meters by moving the decimal 2 places to the left; 85 cm is the same as 0.85 m.

Now that both measurements are in meters, subtract to find out how much taller Ryland is than his brother.

$$\begin{array}{r} 1.60\text{ m} \\ -0.85\text{ m} \\ \hline 0.75\text{ m} \end{array}$$

Ryland is 0.75 meters taller than his brother.

TRY IT 7.105 Mariella is 1.58 meters tall. Her daughter is 75 centimeters tall. How much taller is Mariella than her daughter? Write the answer in centimeters.

TRY IT 7.106 The fence around Hank's yard is 2 meters high. Hank is 96 centimeters tall. How much shorter than the fence is Hank? Write the answer in meters.

EXAMPLE 7.54

Dena's recipe for lentil soup calls for 150 milliliters of olive oil. Dena wants to triple the recipe. How many liters of olive oil will she need?

Solution

We will find the amount of olive oil in milliliters then convert to liters.

	Triple 150 mL
Translate to algebra.	$3 \cdot 150\text{ mL}$
Multiply.	450 mL
Convert to liters.	$450\text{ mL} \cdot \frac{0.001\text{ L}}{1\text{ mL}}$
Simplify.	0.45 L
	Dena needs 0.45 liter of olive oil.

TRY IT 7.107 A recipe for Alfredo sauce calls for 250 milliliters of milk. Renata is making pasta with Alfredo sauce for a big party and needs to multiply the recipe amounts by 8. How many liters of milk will she need?

TRY IT 7.108 To make one pan of baklava, Dorothea needs 400 grams of filo pastry. If Dorothea plans to make 6 pans of baklava, how many kilograms of filo pastry will she need?

Convert Between U.S. and Metric Systems of Measurement

Many measurements in the United States are made in metric units. A drink may come in 2-liter bottles, calcium may come in 500-mg capsules, and we may run a 5-K race. To work easily in both systems, we need to be able to convert between the two systems.

Table 7.4 shows some of the most common conversions.

Conversion Factors Between U.S. and Metric Systems		
Length	**Weight**	**Volume**
1 in = 2.54 cm 1 ft = 0.305 m 1 yd = 0.914 m 1 mi = 1.61 km	1 lb = 0.45 kg 1 oz = 28 g	1 qt = 0.95 L 1 fl oz = 30 mL
1 m = 3.28 ft	1 kg = 2.2 lb	1 L = 1.06 qt

Table 7.4

We make conversions between the systems just as we do within the systems—by multiplying by unit conversion factors.

EXAMPLE 7.55

Lee's water bottle holds 500 mL of water. How many fluid ounces are in the bottle? Round to the nearest tenth of an ounce.

Solution

	500 mL
Multiply by a unit conversion factor relating mL and ounces.	$500 \text{ mL} \cdot \frac{1 \text{ fl oz}}{30 \text{ mL}}$
Simplify.	$\frac{500 \text{ fl oz}}{30}$
Divide.	16.7 fl. oz.
	The water bottle holds 16.7 fluid ounces.

TRY IT 7.109 How many quarts of soda are in a 2-liter bottle?

TRY IT 7.110 How many liters are in 4 quarts of milk?

The conversion factors in Table 7.4 are not exact, but the approximations they give are close enough for everyday purposes. In Example 7.55, we rounded the number of fluid ounces to the nearest tenth.

EXAMPLE 7.56

Soleil lives in Minnesota but often travels in Canada for work. While driving on a Canadian highway, she passes a sign that says the next rest stop is in 100 kilometers. How many miles until the next rest stop? Round your answer to the nearest mile.

Solution

	100 kilometers
Multiply by a unit conversion factor relating kilometers and miles.	$100 \text{ kilometers} \cdot \frac{1 \text{ mile}}{1.61 \text{ kilometers}}$ $100 \cdot \frac{1 \text{ mi}}{1.61 \text{ km}}$
Simplify.	$\frac{100 \text{ mi}}{1.61}$
Divide.	62 mi
	It is about 62 miles to the next rest stop.

TRY IT 7.111 The height of Mount Kilimanjaro is 5,895 meters. Convert the height to feet. Round to the nearest foot.

TRY IT 7.112 The flight distance from New York City to London is 5,586 kilometers. Convert the distance to miles. Round to the nearest mile.

Convert Between Fahrenheit and Celsius Temperatures

Have you ever been in a foreign country and heard the weather forecast? If the forecast is for 22°C. What does that mean?

The U.S. and metric systems use different scales to measure temperature. The U.S. system uses degrees Fahrenheit, written °F. The metric system uses degrees Celsius, written °C. Figure 7.9 shows the relationship between the two systems.

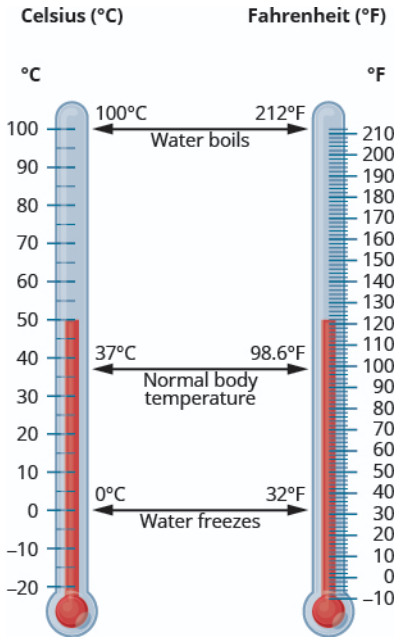


Figure 7.9 A temperature of 37°C is equivalent to 98.6°F.

If we know the temperature in one system, we can use a formula to convert it to the other system.

Temperature Conversion

To convert from Fahrenheit temperature, F, to Celsius temperature, C, use the formula

$$C = \frac{5}{9}(F - 32)$$

To convert from Celsius temperature, C, to Fahrenheit temperature, F, use the formula

$$F = \frac{9}{5}C + 32$$

EXAMPLE 7.57

Convert $50°\text{F}$ into degrees Celsius.

Solution

We will substitute $50°\text{F}$ into the formula to find C.

Use the formula for converting °F to °C	$C = \frac{5}{9}(F - 32)$
Substitute 50 for F.	$C = \frac{5}{9}(50 - 32)$
Simplify in parentheses.	$C = \frac{5}{9}(18)$
Multiply.	$C = 10$
	A temperature of 50°F is equivalent to 10°C.

TRY IT 7.113 Convert the Fahrenheit temperatures to degrees Celsius: $59°\text{F}$.

TRY IT 7.114 Convert the Fahrenheit temperatures to degrees Celsius: $41°\text{F}$.

EXAMPLE 7.58

The weather forecast for Paris predicts a high of $20°\text{C}$. Convert the temperature into degrees Fahrenheit.

Solution

We will substitute $20°\text{C}$ into the formula to find F.

Use the formula for converting °F to °C	$F = \frac{9}{5}C + 32$
Substitute 20 for C.	$F = \frac{9}{5}(20) + 32$
Multiply.	$F = 36 + 32$
Add.	$F = 68$
	So 20°C is equivalent to 68°F.

TRY IT 7.115 Convert the Celsius temperatures to degrees Fahrenheit:

The temperature in Helsinki, Finland was $15°C$.

TRY IT 7.116 Convert the Celsius temperatures to degrees Fahrenheit:

The temperature in Sydney, Australia was $10°C$.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

American Unit Conversion (https://www.openstax.org/l/24USConversion)
Time Conversions (https://www.openstax.org/l/24TimeConversio)
Metric Unit Conversions (https://www.openstax.org/l/24MetricConvers)
American and Metric Conversions (https://www.openstax.org/l/24UStoMetric)
Convert from Celsius to Fahrenheit (https://www.openstax.org/l/24CtoFdegrees)
Convert from Fahrenheit to Celsius (https://www.openstax.org/l/24FtoCdegrees)

SECTION 7.5 EXERCISES

Practice Makes Perfect

Make Unit Conversions in the U.S. System

In the following exercises, convert the units.

214. A park bench is 6 feet long. Convert the length to inches.

215. A floor tile is 2 feet wide. Convert the width to inches.

216. A ribbon is 18 inches long. Convert the length to feet.

217. Carson is 45 inches tall. Convert his height to feet.

218. Jon is 6 feet 4 inches tall. Convert his height to inches.

219. Faye is 4 feet 10 inches tall. Convert her height to inches.

220. A football field is 160 feet wide. Convert the width to yards.

221. On a baseball diamond, the distance from home plate to first base is 30 yards. Convert the distance to feet.

222. Ulises lives 1.5 miles from school. Convert the distance to feet.

223. Denver, Colorado, is $5{,}183$ feet above sea level. Convert the height to miles.

224. A killer whale weighs 4.6 tons. Convert the weight to pounds.

225. Blue whales can weigh as much as 150 tons. Convert the weight to pounds.

226. An empty bus weighs $35{,}000$ pounds. Convert the weight to tons.

227. At take-off, an airplane weighs $220{,}000$ pounds. Convert the weight to tons.

228. The voyage of the *Mayflower* took 2 months and 5 days. Convert the time to days (30 days = 1 month).

229. Lynn's cruise lasted 6 days and 18 hours. Convert the time to hours.

230. Rocco waited $1\frac{1}{2}$ hours for his appointment. Convert the time to seconds.

231. Misty's surgery lasted $2\frac{1}{4}$ hours. Convert the time to seconds.

232. How many teaspoons are in a pint?

233. How many tablespoons are in a gallon?

234. JJ's cat, Posy, weighs 14 pounds. Convert her weight to ounces.

235. April's dog, Beans, weighs 8 pounds. Convert his weight to ounces.

236. Baby Preston weighed 7 pounds 3 ounces at birth. Convert his weight to ounces.

237. Baby Audrey weighed 6 pounds 15 ounces at birth. Convert her weight to ounces.

238. Crista will serve 20 cups of juice at her son's party. Convert the volume to gallons.

239. Lance needs 500 cups of water for the runners in a race. Convert the volume to gallons.

Use Mixed Units of Measurement in the U.S. System

In the following exercises, solve and write your answer in mixed units.

240. Eli caught three fish. The weights of the fish were 2 pounds 4 ounces, 1 pound 11 ounces, and 4 pounds 14 ounces. What was the total weight of the three fish?

241. Judy bought 1 pound 6 ounces of almonds, 2 pounds 3 ounces of walnuts, and 8 ounces of cashews. What was the total weight of the nuts?

242. One day Anya kept track of the number of minutes she spent driving. She recorded trips of $45, 10, 8, 65, 20,$ and 35 minutes. How much time (in hours and minutes) did Anya spend driving?

243. Last year Eric went on 6 business trips. The number of days of each was $5, 2, 8, 12, 6,$ and 3. How much time (in weeks and days) did Eric spend on business trips last year?

244. Renee attached a 6-foot-6-inch extension cord to her computer's 3-foot-8-inch power cord. What was the total length of the cords?

245. Fawzi's SUV is 6 feet 4 inches tall. If he puts a 2-foot-10-inch box on top of his SUV, what is the total height of the SUV and the box?

246. Leilani wants to make 8 placemats. For each placemat she needs 18 inches of fabric. How many yards of fabric will she need for the 8 placemats?

247. Mireille needs to cut 24 inches of ribbon for each of the 12 girls in her dance class. How many yards of ribbon will she need altogether?

Make Unit Conversions in the Metric System

In the following exercises, convert the units.

248. Ghalib ran 5 kilometers. Convert the length to meters.

249. Kitaka hiked 8 kilometers. Convert the length to meters.

250. Estrella is 1.55 meters tall. Convert her height to centimeters.

251. The width of the wading pool is 2.45 meters. Convert the width to centimeters.

252. Mount Whitney is $3{,}072$ meters tall. Convert the height to kilometers.

253. The depth of the Mariana Trench is $10{,}911$ meters. Convert the depth to kilometers.

254. June's multivitamin contains $1{,}500$ milligrams of calcium. Convert this to grams.

255. A typical ruby-throated hummingbird weights 3 grams. Convert this to milligrams.

256. One stick of butter contains 91.6 grams of fat. Convert this to milligrams.

257. One serving of gourmet ice cream has 25 grams of fat. Convert this to milligrams.

258. The maximum mass of an airmail letter is 2 kilograms. Convert this to grams.

259. Dimitri's daughter weighed 3.8 kilograms at birth. Convert this to grams.

260. A bottle of wine contained 750 milliliters. Convert this to liters.

261. A bottle of medicine contained 300 milliliters. Convert this to liters.

Use Mixed Units of Measurement in the Metric System

In the following exercises, solve and write your answer in mixed units.

262. Matthias is 1.8 meters tall. His son is 89 centimeters tall. How much taller, in centimeters, is Matthias than his son?

263. Stavros is 1.6 meters tall. His sister is 95 centimeters tall. How much taller, in centimeters, is Stavros than his sister?

264. A typical dove weighs 345 grams. A typical duck weighs 1.2 kilograms. What is the difference, in grams, of the weights of a duck and a dove?

265. Concetta had a 2-kilogram bag of flour. She used 180 grams of flour to make biscotti. How many kilograms of flour are left in the bag?

266. Harry mailed 5 packages that weighed 420 grams each. What was the total weight of the packages in kilograms?

267. One glass of orange juice provides 560 milligrams of potassium. Linda drinks one glass of orange juice every morning. How many grams of potassium does Linda get from her orange juice in 30 days?

268. Jonas drinks 200 milliliters of water 8 times a day. How many liters of water does Jonas drink in a day?

269. One serving of whole grain sandwich bread provides 6 grams of protein. How many milligrams of protein are provided by 7 servings of whole grain sandwich bread?

Convert Between U.S. and Metric Systems

In the following exercises, make the unit conversions. Round to the nearest tenth.

270. Bill is 75 inches tall. Convert his height to centimeters.

271. Frankie is 42 inches tall. Convert his height to centimeters.

272. Marcus passed a football 24 yards. Convert the pass length to meters.

273. Connie bought 9 yards of fabric to make drapes. Convert the fabric length to meters.

274. Each American throws out an average of $1{,}650$ pounds of garbage per year. Convert this weight to kilograms (2.20 pounds = 1 kilogram).

275. An average American will throw away $90{,}000$ pounds of trash over his or her lifetime. Convert this weight to kilograms (2.20 pounds = 1 kilogram).

276. A 5K run is 5 kilometers long. Convert this length to miles.

277. Kathryn is 1.6 meters tall. Convert her height to feet.

278. Dawn's suitcase weighed 20 kilograms. Convert the weight to pounds.

279. Jackson's backpack weighs 15 kilograms. Convert the weight to pounds.

280. Ozzie put 14 gallons of gas in his truck. Convert the volume to liters.

281. Bernard bought 8 gallons of paint. Convert the volume to liters.

Convert between Fahrenheit and Celsius

In the following exercises, convert the Fahrenheit temperature to degrees Celsius. Round to the nearest tenth.

282. $86°F$ **283**. $77°F$ **284**. $104°F$

285. $14°F$ **286**. $72°F$ **287**. $4°F$

288. $0°F$ **289**. $120°F$

In the following exercises, convert the Celsius temperatures to degrees Fahrenheit. Round to the nearest tenth.

290. $5°C$ **291**. $25°C$ **292**. $-10°C$

293. $-15°C$ **294**. $22°C$ **295**. $8°C$

296. $43°C$ **297**. $16°C$

Everyday Math

298. **Nutrition** Julian drinks one can of soda every day. Each can of soda contains 40 grams of sugar. How many kilograms of sugar does Julian get from soda in 1 year?

299. **Reflectors** The reflectors in each lane-marking stripe on a highway are spaced 16 yards apart. How many reflectors are needed for a one-mile-long stretch of highway?

Writing Exercises

300. Some people think that $65°$ to $75°$ Fahrenheit is the ideal temperature range.
ⓐ What is your ideal temperature range? Why do you think so?

ⓑ Convert your ideal temperatures from Fahrenheit to Celsius.

301. ⓐ Did you grow up using the U.S. customary or the metric system of measurement? ⓑ Describe two examples in your life when you had to convert between systems of measurement. ⓒ Which system do you think is easier to use? Explain.

Self Check

ⓐ After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.

I can...	Confidently	With some help	No-I don't get it!
make unit conversions in the U.S. system.			
use mixed units of measurement in the U.S. system.			
use mixed units of measurement in the metric system.			
convert between the U.S. and the metric systems of measurement.			
convert between Fahrenheit and Celsius temperatures.			

ⓑ Overall, after looking at the checklist, do you think you are well-prepared for the next chapter? Why or why not?